

Biology

Q. 2

i)

It was hypothesized that mosquitoes spread malaria from one person to another.

ii)

Ronald Ross allowed mosquitoes to feed on infected sparrows and then allowed those mosquitoes to bite healthy sparrows.

The healthy sparrows developed malaria, proving that mosquitoes could transmit the disease.

iii)

The hypothesis was tested by allowing mosquitoes that had fed on malaria patients to bite healthy people. The people then developed malaria, confirming mosquito transmission.

Q. 23

i)

Bacteria

They are prokaryotic cells and lack a nucleus.

Protists

They are eukaryotic cells and have a nucleus.

② They generally lack membrane bound organelles. They possess membrane bound organelles.

organelles

ii)

Fungi

It cannot make their own food and obtain nutrients from other sources.

Their cell walls are mainly made of chitin.

Plants

It makes their own food by process called photosynthesis.

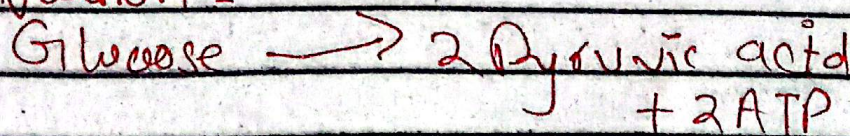
Their cell walls are mainly made of cellulose.

Q: 7

(i) Glycolysis takes place in cytoplasm.

One glucose molecule is converted into 2 pyruvic acid molecules, producing a net gain of 2 ATP.

Equation:-



(i)

In order :-

Glycolysis

Link reaction

Krebs cycle.

Electron transport chain.

Q.4

(i)

functions:-

① Lipid synthesis

② Detoxification of drugs & harmful substances.

(ii) Salivary gland cell :-

Two abundant organelles are :-

* Ribosomes - for making proteins / enzymes.

* RER (rough endoplasmic reticulum) - for synthesis and transport of proteins.

Q. 5

i) Xylem

It transports water and mineral salts from roots to other parts

Mainly consists of dead cell.

Phloem

It transports food/sugars from source to sinks.

Mainly consists of living cell.

ii) Dermal tissue

forms the outer protective covering of the plant.

Protects against physical damage and water loss.

Ground tissue

Makes up the tissue b/w dermal and vascular tissues.

Performs functions such as photosynthesis, storage and support

Q. 6

i) Carbon : Hydrogen : Oxygen
1 : 2 : 1

General formula:-
 $(CH_2O)_n$

ii) chromatin :- is composed of DNA
 & proteins, mainly histones

When DNA coils around histone
 protein, it forms bead-like
 structure called nucleosomes.

Q:8

i)

A : H_2O

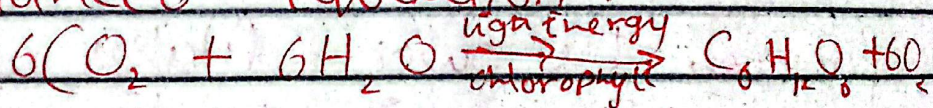
B : CO_2

C : O_2

D : Sugar

ii)

Balanced Equation :-



iii)

Energy molecules

1. ATP

2. NADP

Q.9
(i)

meiosis I

Homologous chromosomes separate

2. Produces 2 haploid cells

(3) Crossing over occurs in prophase I.

(4) Chromosome number is reduced by half.

ii)

(a) = 9 chromosome

(b) 9

(c) 4

iii)

By crossing over and independent produce different combinations of genes / chromosome into

meiosis II

Sister chromatids separate.

(1) Produces four haploid cells in total.

Crossing over does not occur.

Chromosomes n.o remains the same

daughter cells.

∴ M(C)S:

1. B.
2. C
3. A
4. D
5. A
6. A
7. D
8. C
9. A
10. B